

Part VIII — Electricity and Magnetism · Chapter 40

Electric Charge

The fundamental property that creates electric forces, fields, and all of electromagnetic behaviour

Shuffle across a carpet and touch a doorknob: you feel a sharp spark. Rub a balloon on your hair and it sticks to the wall. Comb dry hair and small bits of paper leap toward the comb. Pull two pieces of plastic wrap apart and they cling stubbornly to everything. These are all manifestations of electric charge — a fundamental property of matter just as mass is a fundamental property. While mass creates gravitational attraction, charge creates electric attraction and repulsion. Electricity, magnetism, light, chemistry, and the cohesion of atoms themselves all arise from electric charge and the forces between charges. This chapter introduces what charge is, how it is transferred, and how charges exert forces on each other.

40.1 What Electric Charge Is

Electric charge is a fundamental intrinsic property of certain subatomic particles. Like mass, it is not explained by any more fundamental property — it simply is. Unlike mass, charge comes in two varieties (positive and negative) and can cancel: equal amounts of positive and negative charge in the same region produce no net charge effect.

Charge is quantised: it comes only in integer multiples of the elementary charge $e = 1.602 \times 10^{-19} \text{ C}$. The SI unit of charge is the coulomb (C). One coulomb is an enormous amount of charge — the charge on about 6.24×10^{18} elementary particles. The charge of a single electron is $-e = -1.602 \times 10^{-19} \text{ C}$; a proton carries $+e = +1.602 \times 10^{-19} \text{ C}$. No free particle with a charge smaller than e has ever been observed (quarks have fractional charge, but they are always confined in groups whose total charge is an integer multiple of e).

Electric charge creates an electric force. Two charged objects exert forces on each other: like charges (both positive, or both negative) repel; unlike charges (one positive, one negative) attract. The strength of this force depends on the amount of charge and the distance between the charges, following Coulomb's Law (Section 40.10).

Key Point: Electric charge is analogous to mass in classical mechanics: both are fundamental properties of matter, both create forces (gravitational for mass; electric for charge), and both are conserved. But charge can be negative, creating repulsion as well as attraction, while mass is always positive and gravity is always attractive.

40.2 Positive and Negative Charge

The labelling of charge as “positive” and “negative” is a convention introduced by Benjamin Franklin in the 18th century. Franklin did not know about electrons and protons; he simply observed that there were two kinds of charge that cancelled each other. He chose to call the charge acquired by a glass rod rubbed with silk “positive” and the charge acquired by rubber rubbed with fur “negative.” We still use this convention.

The choice of sign for which type of particle to call positive was arbitrary, but it has consequences: in a metal wire carrying conventional current, current is defined as the direction positive charges would flow, which is opposite to the direction electrons (the actual charge carriers) move. This historical accident occasionally causes confusion but is deeply embedded in all engineering and physics convention.

The key physical rules are: like charges repel, unlike charges attract. Two positive charges push each other apart; two negative charges push each other apart; a positive and a negative charge pull toward each other. The magnitude of this force depends on how much charge is present and the distance between them.

40.3 Protons, Electrons, and Ions

In ordinary matter, the charge of an atom is determined by the numbers of protons (positive, in the nucleus) and electrons (negative, in orbitals around the nucleus). A neutral atom has equal numbers of protons and electrons, giving zero net charge. The nuclear charge of a proton is $+1.602 \times 10^{-19}$ C; each electron has charge -1.602×10^{-19} C.

If an atom gains one or more electrons, it has more negative charge than positive and becomes a negative ion (anion). If it loses electrons, it becomes a positive ion (cation). Chemical bonds, electrochemistry, and the behaviour of salts and acids in solution all involve the transfer and sharing of electrons between atoms. When we say an object has been charged electrically, we mean that electrons have been added to or removed from it — protons, being locked in the nucleus, do not move in ordinary chemistry or contact electricity.

Particle	Location	Charge	Mass
Proton	Nucleus	$+1.602 \times 10^{-19} \text{ C (+e)}$	$1.673 \times 10^{-27} \text{ kg}$
Neutron	Nucleus	0	$1.675 \times 10^{-27} \text{ kg}$
Electron	Orbitals	$-1.602 \times 10^{-19} \text{ C (-e)}$	$9.109 \times 10^{-31} \text{ kg}$

Table 40.1 *Properties of the subatomic particles that make up ordinary matter. Electrons are the particles that move when an object is electrically charged.*

The mass of an electron is about 1/1836 the mass of a proton. Despite this tiny mass, the electron's charge is equal in magnitude to the proton's. The electron's mobility is what makes electrical phenomena in solids possible: electrons can be displaced by external electric fields while the much heavier protons remain fixed in the crystal lattice.

40.4 Conductors and Insulators

Materials differ enormously in their ability to allow charge to move through them. This difference determines almost everything about how they behave electrically.

Conductors

A conductor is a material in which electrons can move freely throughout the material. Metals are the most important class of conductors: in a metal, the outer electrons of each atom are not tightly bound and form a "sea" of mobile electrons that can flow throughout the material. When an external electric force is applied (for example, by a battery), these free electrons drift, creating an electric current. Common conductors include copper, aluminium, silver, gold, iron, and saltwater.

Insulators

An insulator (or dielectric) is a material in which electrons are tightly bound to individual atoms or molecules and cannot flow freely. Applying an electric force moves electrons only tiny distances from their equilibrium positions before restoring forces balance the applied force. Charge placed on an insulator stays where it is placed rather than spreading over the surface. Common insulators include glass, plastic (polyethylene, PVC), rubber, dry wood, and air.

Semiconductors

Semiconductors (silicon, germanium, gallium arsenide) have electrical conductivity between metals and insulators. Their conductivity can be altered by adding small amounts of impurities (doping) or by applying heat, light, or electric fields. This tunability makes semiconductors the basis of all modern electronics: transistors, integrated circuits, LEDs, solar cells, and computer chips.

Superconductors

Superconductors are materials that, below a critical temperature, allow electric current to flow with literally zero electrical resistance. The charge carriers in a superconductor form quantum-mechanical “Cooper pairs” that propagate without scattering. Superconductors have applications in MRI machines (superconducting magnets), particle accelerators, and are the subject of intense research for energy transmission.

Class	Example materials	Free electrons?	Conductivity (S/m)
Conductor	Copper, silver, aluminium	Yes — mobile electron sea	$\sim 10^6\text{--}10^8$
Semiconductor	Silicon, germanium	Few; tuneable by doping	$\sim 10^{-3}\text{--}10^3$
Insulator	Glass, rubber, plastic	Essentially none	$\sim 10^{-13}\text{--}10^{-10}$
Superconductor	Niobium, YBCO (cooled)	Paired (zero resistance)	Effectively infinite

Table 40.2 *Categories of materials by electrical conductivity. The range spans over 20 orders of magnitude from best insulator to best conductor.*

40.5 Charging by Friction

When two different materials are rubbed together, electrons transfer from one material to the other. The material that tends to give up electrons becomes positively charged; the material that gains electrons becomes negatively charged. The charge acquired by each material is equal in magnitude and opposite in sign — total charge is conserved.

The triboelectric series ranks materials by their tendency to give up electrons (positive

end) or gain electrons (negative end) when rubbed:

Positive end (loses electrons): human skin, glass, hair, wool, nylon, lead, silk, aluminium, paper — neutral materials in the middle — cotton, steel, wood, amber, rubber, polyethylene (negative end: gains electrons).

When you rub a glass rod with silk, the glass loses electrons to the silk. The glass rod becomes positively charged and the silk negatively charged. When you rub rubber with fur, the rubber gains electrons and becomes negatively charged. The carpet-and-shoe combination that causes static shocks involves your shoe soles (rubber or synthetic) gaining electrons from the carpet fibres.

Historical Note: Benjamin Franklin (1706–1790) demonstrated in 1752 that lightning is an electrical discharge using his famous (and extremely dangerous) kite experiment. He proposed the single-fluid model of electricity and introduced the terms “positive” and “negative.” Franklin also invented the lightning rod, saving countless buildings from fire by providing a low-resistance path to ground for lightning strikes.

40.6 Charging by Contact

A charged object can transfer charge to a neutral conductor by direct contact. When a negatively charged rod is touched to a neutral metal sphere, electrons flow from the rod onto the sphere (since like charges repel and electrons spread to minimise repulsion). The sphere acquires a net negative charge. When contact is broken, the sphere retains this charge.

The amount of charge transferred by contact depends on the sizes and geometries of the objects and the initial charge. In general, charge distributes itself so that the potential is equalised between the two touching conductors — a concept we will formalise in Chapter 41. After contact, each object may have different total charge, but the potential at their surfaces is the same.

An important special case is grounding: connecting an object to the Earth. Because the Earth is an enormous conductor, it acts as an infinite charge reservoir. Grounding a positively charged sphere allows electrons from the Earth to flow onto the sphere, neutralising it. Grounding a negatively charged sphere allows its excess electrons to flow into the Earth. After grounding, the object is neutral. This is why electrical grounding is essential for safety: it prevents dangerous charge build-up on equipment and provides a path for fault currents.

40.7 Charging by Induction

Induction charges an object without direct contact by rearranging the charges already in it. The mechanism relies on the fact that conductors have free electrons that can move in response to external electric forces.

Process for charging a conductor by induction:

Step 1: Approach: Bring a charged object (say, negatively charged) near but not touching a neutral conducting sphere.

Step 2: Redistribution: The charged object's electric field repels free electrons in the sphere. Electrons migrate to the far side of the sphere. The near side of the sphere becomes positively charged (electron deficit); the far side becomes negatively charged (electron surplus). The sphere as a whole is still neutral but has separated charge.

Step 3: Ground: While the charged object is still near, touch the sphere briefly with a finger or ground wire. Electrons repelled to the far side escape to ground (or to your body). The near side retains its positive charge.

Step 4: Remove ground: Break the ground connection first, before removing the charged rod.

Step 5: Remove rod: Remove the original charged rod. The positive charge on the sphere redistributes uniformly. The sphere is now positively charged without ever having touched the charged rod.

Notice: the charged rod that did the inducing retains its original charge throughout. Charging by induction does not discharge the inducing object. Also notice: the sign of the induced charge is opposite to the sign of the inducing charge (the sphere becomes positive when approached by a negative rod).

Key Point: Induction differs from contact charging in two important ways: (1) no charge is transferred between the two objects; and (2) the induced charge is opposite in sign to the inducing charge, while contact charging produces charge of the same sign as the charging object.

40.8 Conservation of Charge

Conservation of charge is one of the most fundamental and precisely tested laws of physics: the total electric charge of an isolated system never changes. Charge can be transferred between objects, separated, or recombined, but it cannot be created from nothing or destroyed.

$$\Delta Q_{\text{total}} = 0 \quad (\text{isolated system})$$

Conservation of electric charge. The total charge of an isolated system is constant. Charge can be

moved around but never created or destroyed.

When a glass rod is rubbed with silk, the positive charge acquired by the glass is exactly equal in magnitude to the negative charge acquired by the silk. When an electron and a positron (antielectron) annihilate, they produce photons (which carry no charge) — the total charge before ($-1e + 1e = 0$) equals the total charge after (0). In every nuclear reaction, decay, and particle collision ever observed, the total charge is conserved.

Conservation of charge is a consequence of a deep symmetry in the laws of physics called gauge invariance. Unlike conservation of energy (which follows from time-translation symmetry) or conservation of momentum (from space-translation symmetry), charge conservation follows from the gauge symmetry of the electromagnetic field. This is a connection to advanced physics, but the practical result is simple and absolute: you cannot create or destroy charge.

The Rule: In any charging process — friction, contact, induction, ionisation, or nuclear reaction — the net charge gained by all objects involved sums to zero. If object A gains charge $+Q$, object B (or the environment) gains charge $-Q$.

40.9 Electric Force Between Charges

Charges exert forces on each other without contact. This “action at a distance” puzzled physicists for centuries. The modern understanding is that each charge creates an electric field in the space around it, and that field exerts force on any other charge placed in it (Chapter 41). For now, we treat the interaction directly.

The key qualitative rules are: like charges repel, unlike charges attract. The force acts along the line connecting the two charges. The force on each charge is equal in magnitude and opposite in direction (Newton’s third law). The force decreases with distance — specifically, it follows an inverse-square law identical in mathematical form to Newton’s law of gravitation.

The force between two point charges was quantified by Charles-Augustin de Coulomb in 1785 through torsion balance experiments, giving Coulomb’s Law.

40.10 Coulomb's Law

Coulomb's Law gives the magnitude of the electric force between two point charges:

$$\mathbf{F} = k |q_1| |q_2| / r^2$$

Coulomb's Law. F = electric force (N); k = Coulomb's constant = $8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$ (often written as $1/(4\pi\epsilon_0)$); q_1, q_2 = charges (C) with magnitudes taken as positive; r = separation between charges (m). The force is repulsive if the charges have the same sign, attractive if opposite.

Coulomb's constant $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$ is an enormous number. Even a microcoulomb (10^{-6} C) of charge produces forces of many newtons at centimetre separations. This is why everyday objects are nearly perfectly charge-neutral: even a tiny charge imbalance produces very strong restoring forces that drive charge to redistribute until balance is achieved.

Coulomb's Law is mathematically identical to Newton's gravitational law ($F = Gm_1m_2/r^2$), with charge replacing mass and k replacing G . But the magnitudes are vastly different: the electric force between a proton and an electron is about $2.3 \times 10^{-28} \text{ N}$, while the gravitational force between them is only $3.6 \times 10^{-47} \text{ N}$ — the electric force is about 10^{131} times stronger. Gravity dominates at astronomical scales only because matter is, on average, almost perfectly electrically neutral.

Historical Note: Charles-Augustin de Coulomb (1736–1806) used a sensitive torsion balance of his own design to measure the forces between charged spheres. By varying the charges and separations systematically, he verified the inverse-square law and the proportionality to each charge. His 1785 papers gave the quantitative law now bearing his name. The unit of charge, the coulomb, was named in his honour in 1881.

Example — Force between two charges Two small balls carry charges $q_1 = +3.0 \mu\text{C}$ and $q_2 = -5.0 \mu\text{C}$ and are 15 cm apart. $F = k|q_1||q_2|/r^2 = (8.99 \times 10^9)(3.0 \times 10^{-6})(5.0 \times 10^{-6})/(0.15)^2 = (8.99 \times 10^9)(15 \times 10^{-12})/(0.0225) = (1.349 \times 10^{-1})/(0.0225) \approx 6.0 \text{ N}$. The force is attractive because the charges have opposite signs.

Example — Comparing electric and gravitational forces Two protons ($m = 1.673 \times 10^{-27} \text{ kg}$, $q = +1.602 \times 10^{-19} \text{ C}$) are $1.0 \times 10^{-10} \text{ m}$ apart (about atomic scale). $F_E = kq^2/r^2 = (8.99 \times 10^9)(1.602 \times 10^{-19})^2/(10^{-10})^2 \approx 23 \text{ N}$. $F_G = Gm^2/r^2 = (6.67 \times 10^{-11})(1.673 \times 10^{-27})^2/(10^{-10})^2 \approx 1.9 \times 10^{-26} \text{ N}$. Ratio: $F_E/F_G \approx 10^{27}$. Electromagnetism dominates gravity at atomic and molecular scales.

40.11 Attraction, Repulsion, and Distance Effects

The inverse-square dependence of Coulomb's Law ($\text{force} \propto 1/r^2$) means the electric force falls off rapidly with distance, but never reaches zero. Doubling the separation reduces the force to one-quarter; tripling it reduces the force to one-ninth. This is the same mathematical form as gravity, and it arises from the same geometric reason: the force spreads over the surface area of a sphere, which grows as r^2 .

Multiple charges exert forces independently. The net force on a charge due to several others is the vector sum of all the individual Coulomb forces. This principle of superposition is exact and allows complex charge distributions to be analysed by summing the contributions of individual charges.

Even neutral objects can be attracted to charged objects. When a charged rod is brought near a neutral conductor, the nearer face develops induced charge of opposite sign (Section 40.7). The attractive force to the near face is stronger than the repulsive force from the far face (which is farther away and therefore the force is weaker). The net result is attraction. This is why a charged comb picks up small pieces of paper, even though the paper carries no net charge.

Example — Superposition of forces Three charges are in a line: $q_1 = +2.0 \mu\text{C}$ at $x = 0$, $q_2 = -1.0 \mu\text{C}$ at $x = 0.30 \text{ m}$, $q_3 = +2.0 \mu\text{C}$ at $x = 0.60 \text{ m}$. Find the net force on q_2 . Force from q_1 on q_2 : $F_{12} = k(2.0 \times 10^{-6})(1.0 \times 10^{-6})/(0.30)^2 = (8.99 \times 10^9)(2 \times 10^{-12})/0.09 \approx 0.20 \text{ N}$, directed toward q_1 (attractive, so in the $-x$ direction). Force from q_3 on q_2 : same magnitude by symmetry, directed toward q_3 (attractive, so in the $+x$ direction). Net force = 0 by symmetry. The middle charge is in equilibrium.

40.12 Electric Charge in Everyday Technology

Electric charge and its manipulation underlie almost all modern technology.

Photocopiers and laser printers use electrostatic attraction: a photosensitive drum is charged uniformly, then selectively discharged by a laser beam (printer) or light image (copier). Charged toner powder sticks to the remaining charged areas of the drum and is then transferred to paper by further electrostatic attraction, then fused by heat.

Electrostatic precipitators remove dust, soot, and particulate matter from industrial flue gases. Particles passing through a high-voltage region acquire charge, then are attracted to and deposited on oppositely charged collector plates. Modern power plants and cement factories use precipitators to remove over 99% of particulates.

Capacitive touchscreens (smartphones, tablets) detect the position of a conducting finger by measuring the change in capacitance at the touch point. The finger, being a conductor, disturbs the electric field maintained between the screen's transparent conducting layers. The device maps these disturbances to screen positions.

Spray painting with electrostatic guns charges paint droplets before they leave the nozzle. The charged droplets are attracted to the grounded object being painted, wrapping around edges and covering surfaces that a direct spray would miss. Electrostatic painting achieves higher transfer efficiency (less wasted paint) and more uniform coverage than conventional spraying.

Van de Graaff generators build up large static charges by mechanically carrying charge on a moving belt to a hollow conducting sphere, which stores the charge on its outer surface (all charge on a conductor resides at the surface). They produce voltages of hundreds of thousands of volts and are used in physics demonstrations and as particle accelerators for nuclear research.

40.13 Common Mistakes in Charge Problems

- Forgetting to convert microcoulombs to coulombs. Charges are often given in microcoulombs ($\mu\text{C} = 10^{-6} \text{ C}$) or nanocoulombs ($\text{nC} = 10^{-9} \text{ C}$). Substituting these directly into Coulomb's Law without converting gives answers off by factors of 10^6 or 10^9 . Always convert to SI (coulombs) before calculating.
- Confusing the direction of force. Coulomb's Law gives only the magnitude. You must determine direction separately from the rule: like charges repel (force points away from the other charge); unlike charges attract (force points toward the other charge). In superposition problems, assign positive and negative signs to forces in your chosen coordinate direction before summing.
- Applying Coulomb's Law to non-point charges incorrectly. Coulomb's Law applies to point charges and to uniformly charged spheres (where all charge can be treated as concentrated at the centre). For other extended charge distributions, integration or superposition of differential elements is required.
- Thinking that inducing a charge transfers charge from the inducing object. Induction does not transfer charge from the inducing rod to the object. The rod retains its original charge throughout. Only contact charging transfers charge from one object to another.
- Incorrectly applying conservation of charge in contact problems. When two identically sized conducting spheres touch, the total charge distributes equally. But if one sphere is larger, charge distributes in proportion to capacitance (roughly surface area). For identical spheres: each gets $(q_1 + q_2)/2$ after contact.
- Forgetting that Coulomb forces are vectors, not scalars. In problems with three or more charges, compute each Coulomb force as a vector (magnitude and

direction), then add the vectors. Simply adding magnitudes ignores direction and gives wrong answers whenever forces are not all in the same direction.

Common Mistake: Always convert charge units to coulombs before using Coulomb's Law. A charge of $3.0 \mu\text{C}$ is $3.0 \times 10^{-6} \text{ C}$, not 3.0 C . Substituting 3.0 C instead of $3.0 \times 10^{-6} \text{ C}$ gives an answer 10^{12} times too large — a physically impossible result.

Chapter Summary

Electric charge is a fundamental property of matter. Charge comes in positive and negative varieties; like charges repel and unlike charges attract. Charge is quantised in units of $e = 1.602 \times 10^{-19} \text{ C}$ and is absolutely conserved.

- Electrons carry charge $-e$; protons carry $+e$. Ordinary charging involves electron transfer; protons stay in the nucleus.
- Conductors allow free electron movement; insulators do not. Semiconductors are intermediate and tuneable.
- Charging methods: friction (triboelectric transfer), contact (charge flows between touching objects), induction (charge redistributed by external field without contact; sign opposite to inducer).
- Conservation of charge: total charge of an isolated system is constant. Net charge created = 0 in any process.
- Coulomb's Law: $F = k|q_1||q_2|/r^2$; $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$. Inverse-square law, identical in form to gravity.
- Electric force is 10^{36} times stronger than gravity at atomic scale. Gravity dominates large scales only because matter is nearly charge-neutral.
- Superposition: net force on a charge is the vector sum of all individual Coulomb forces from other charges.

Key Terms

electric charge: A fundamental intrinsic property of matter; comes in positive and negative varieties; SI unit coulomb (C).

elementary charge (e): $e = 1.602 \times 10^{-19}$ C; the magnitude of charge on one proton or one electron; the minimum observable free charge.

coulomb (C): SI unit of electric charge; $1\text{ C} =$ charge of about 6.24×10^{18} protons.

conductor: A material with free electrons that allow charge to flow readily; metals are primary examples.

insulator: A material with no free electrons; charge cannot flow; examples: glass, rubber, plastic.

semiconductor: A material with intermediate, tuneable conductivity; basis of modern electronics.

grounding: Connecting an object to Earth (an effectively infinite charge reservoir) to neutralise excess charge.

charging by friction: Transfer of electrons between two different materials rubbed together; produces equal and opposite charges.

charging by contact: Transfer of charge from a charged object to a neutral conductor through direct physical contact.

charging by induction: Redistribution of charge in a conductor caused by a nearby charged object, without direct contact; induced charge is opposite in sign to inducing charge.

conservation of charge: The total electric charge of an isolated system is constant; charge cannot be created or destroyed.

Coulomb's Law: $F = k|q_1||q_2|/r^2$; the magnitude of the electric force between two point charges separated by distance r .

Coulomb's constant (k): $k = 8.99 \times 10^9\text{ N}\cdot\text{m}^2/\text{C}^2$; proportionality constant in Coulomb's Law.

triboelectric series: A ranking of materials by their tendency to give up or gain electrons when rubbed; determines the sign and magnitude of charging by friction.

superposition principle: The net electric force on a charge is the vector sum of the individual forces exerted by all other charges.

Concept Review Questions

Answer in complete sentences. No calculations unless stated.

1. Explain why rubbing a glass rod with silk charges both objects. Where do the charges come from? Does the total charge in the room change?
 2. A copper rod held in your bare hand cannot be charged by rubbing. Explain why, and describe what you would need to do differently to charge the copper rod.
 3. Describe the three-step process for charging a metal sphere by induction. Why must the ground connection be broken before the inducing rod is removed?
 4. A charged balloon sticks to a neutral wall. Use the concept of charge induction to explain why. Does the wall need to be a conductor for this to work?
 5. Coulomb's Law and Newton's Law of Gravitation have the same mathematical form. List two ways they are mathematically analogous and two important physical differences.
 6. The electric force between a proton and an electron is about 10^{36} times stronger than the gravitational force between them at the same distance. Why, then, does gravity dominate the large-scale structure of the universe?
 7. Explain the difference between charging by contact and charging by induction in terms of (a) whether the inducing object loses charge, and (b) the sign of the charge acquired.
 8. Two identical conducting spheres carry charges $+8\ \mu\text{C}$ and $-2\ \mu\text{C}$. They are touched together and then separated. What charge does each sphere now carry? Justify your answer using conservation of charge.
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Charge Interaction Exercises

9. Two charged pith balls hang from threads 30 cm long. They repel each other and hang at an angle such that they are separated by 4.0 cm. If each ball has

- mass 0.20 g and charge q , what is the approximate charge on each? (Hint: the tension, weight, and electric force form a triangle of forces.)
10. Three charges are on the x-axis: $+4.0\ \mu\text{C}$ at $x = 0$, $-2.0\ \mu\text{C}$ at $x = 0.20\ \text{m}$, and $+3.0\ \mu\text{C}$ at $x = 0.40\ \text{m}$. Find the net force on the middle charge (magnitude and direction).
 11. A neutral conducting sphere is approached by a rod carrying $-20\ \mu\text{C}$. Describe the distribution of charge on the sphere. Does the sphere experience a net force? Explain.
 12. A glass rod rubbed with silk acquires charge $+120\ \text{nC}$. How many electrons were transferred, and in which direction — from silk to glass or from glass to silk?
 13. Two identical conducting spheres carry $+6\ \mu\text{C}$ and $-10\ \mu\text{C}$ respectively. (a) Find the force between them when 20 cm apart. (b) The spheres are touched and separated. Find the new force between them. Is the force attractive or repulsive now?
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Coulomb's Law Problems

14. Find the distance at which the electric force between two $+1.0\ \text{C}$ charges equals the weight of a 70 kg person. ($F = mg = 686\ \text{N}$) Comment on whether 1 C is a large or small amount of charge.
 15. An electron and proton in a hydrogen atom are $5.3 \times 10^{-11}\ \text{m}$ apart. (a) Find the electric force. (b) Find the gravitational force. (c) Find the ratio F_E/F_G .
 16. Three charges form a right triangle. $q_1 = +3.0\ \mu\text{C}$ is at the origin, $q_2 = -4.0\ \mu\text{C}$ is at $(0.30\ \text{m}, 0)$, and $q_3 = +2.0\ \mu\text{C}$ is at $(0, 0.40\ \text{m})$. Find the magnitude and direction of the net force on q_3 .
 17. Two small spheres of mass 5.0 g each carry identical positive charges q . They are suspended from the same point by threads 0.50 m long and hang in equilibrium separated by 0.15 m. Find q .
 18. At what point on the line between a $+9.0\ \mu\text{C}$ charge and a $+4.0\ \mu\text{C}$ charge (separated by 1.0 m) is the net force on a test charge zero? (Set up and solve the equation; the answer is not the midpoint.)
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Charging Method Analysis

19. A student charges a metal rod by (a) holding it in bare hands and rubbing it with wool; (b) holding it with rubber-handled pliers and rubbing it with wool; (c) holding it with rubber-handled pliers, touching it briefly to a charged sphere, then separating. For each method, predict whether the rod will be charged and explain why or why not.
20. Describe what happens, step by step, when a positively charged rod is brought near a neutral electroscope (a device with thin metal leaves that spread apart when charged). The electroscope is then grounded while the rod is still near, then the ground is removed, then the rod is taken away. What charge does the electroscope have at the end?
21. Two conducting spheres A (radius r) and B (radius $3r$) are connected by a wire. A charge of $80\text{ }\mu\text{C}$ is placed on sphere A. When equilibrium is reached, the charge distributes between the spheres in proportion to their radii (for conducting spheres: $q \propto \text{radius}$). How much charge ends up on each sphere? (Hint: $q_A/q_B = r_A/r_B$ and $q_A + q_B = 80\text{ }\mu\text{C}$.)
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Applied Electrostatics Problems

22. An electrostatic precipitator must remove soot particles (typical charge $50e$ per particle) from a flue gas stream. An electric force of at least $1.5 \times 10^{-14}\text{ N}$ must act on each particle to deflect it to the collector plate within the precipitator. If the collecting electrode creates a field that applies this force, and the field acts

- over a separation of 0.10 m, find the voltage difference between the electrodes ($V = F \times d / q$). Does this seem like a reasonable industrial voltage?
23. A Van de Graaff generator builds up a charge of $+15 \mu\text{C}$ on its dome (radius 0.20 m). Model the dome as a point charge at its centre. (a) Find the force on a $+1.0 \mu\text{C}$ test charge placed 0.50 m from the dome centre. (b) If a person with net charge $-0.5 \mu\text{C}$ stands 1.5 m from the dome, find the force on them. Is this likely to be perceptible?
24. Two charges ($+5 \mu\text{C}$ and $-5 \mu\text{C}$) are placed 0.10 m apart, forming an electric dipole. Find the force on a $+2.0 \mu\text{C}$ charge placed (a) on the line joining the two charges, 0.30 m from the positive charge (so 0.40 m from the negative charge); (b) at the midpoint of the dipole line. Which position gives a larger net force, and why?
25. An ink-jet printer works by charging ink droplets and deflecting them with an electric field. A droplet of mass $1.0 \times 10^{-13} \text{ kg}$ must be deflected 2.5 mm sideways while travelling 15 mm horizontally through the deflection region. If the droplet travels at 20 m/s and experiences a transverse electric force F , find the required F . If the deflecting plates are 5.0 mm apart and the voltage between them is V , find V (using $F = qV/d$ and assuming the droplet carries charge $q = 8 \times 10^{-17} \text{ C}$).

Chapter 40 Checkpoint

Before moving on to Chapter 41, confirm you can do each of the following.

I can...	See Section
Define electric charge and state its SI unit and quantisation	40.1
Explain positive and negative charge and the historical convention	40.2
State the charges and masses of protons, electrons, and neutrons	40.3
Distinguish conductors, insulators, and semiconductors	40.4
Explain charging by friction and use the triboelectric series qualitatively	40.5
Describe charging by contact and explain grounding	40.6
Describe charging by induction in five steps	40.7
State and apply conservation of charge to charging problems	40.8
Apply Coulomb's Law to calculate force between two point charges	40.10
Apply the superposition principle to find force from multiple charges	40.11
Identify common mistakes in charge calculations and avoid them	40.13

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